

Emotion in Meaning: Cross-Linguistic Effects of Valence and Arousal on Semantic Judgments

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Research Questions

Q1: Do valence **and** arousal interact to affect semantic processing speed and accuracy? **Is there a Valence-Arousal Conflict** (e.g., Robinson et al., 2004)?

Q2: Can this be observed across both the first language (L1) and second language (L2) for bilingual speakers? **Do bilinguals process emotions differently across their two languages?**

Method

Word list **matched across languages** (word frequency, length, orthographic neighbourhood density, cognate status)

Affective **judgement task** (positive/negative and arousing/calming)

Two different bilingual groups (differing on immersion)

Four experiments

Key Findings

Early emotional processing in bilinguals may reflect **automatic cognitive mechanisms** rather than language-specific effects.

Arousal had a stronger effect (compared to valence) on semantic processing

We found **evidence for a Valence-Arousal Conflict**: high arousal negative and low arousal positive stimuli is processed more efficiently

*However, **only** for the arousal task*

May be due to interactions with word frequency at earlier processing window

Bilingual demographics had a moderating influence on both valence and arousal (e.g., age of acquisition, proficiency, immersion, and language use)

Arousing				Calming			
Negative/unpleasant				Positive/pleasant			
Low arousal		High arousal		Low arousal		High arousal	
unförmig	deformed	Spinne	spider	Straße	street	bewundert	admired
dunkel	dark	Angriff	assault	Fenster	window	Jubel	joy
Harn	urine	demütigen	humiliate	Baum	tree	Leidenschaft	passion
trüb	muddy	Geisel	hostage	ernähren	nourish	Geld	money
starr	rigid	Kakerlake	cockroach	sauber	tidy	strahlend	radiant



Exp 1:

Valence Task



Exp 2:



Exp 3:

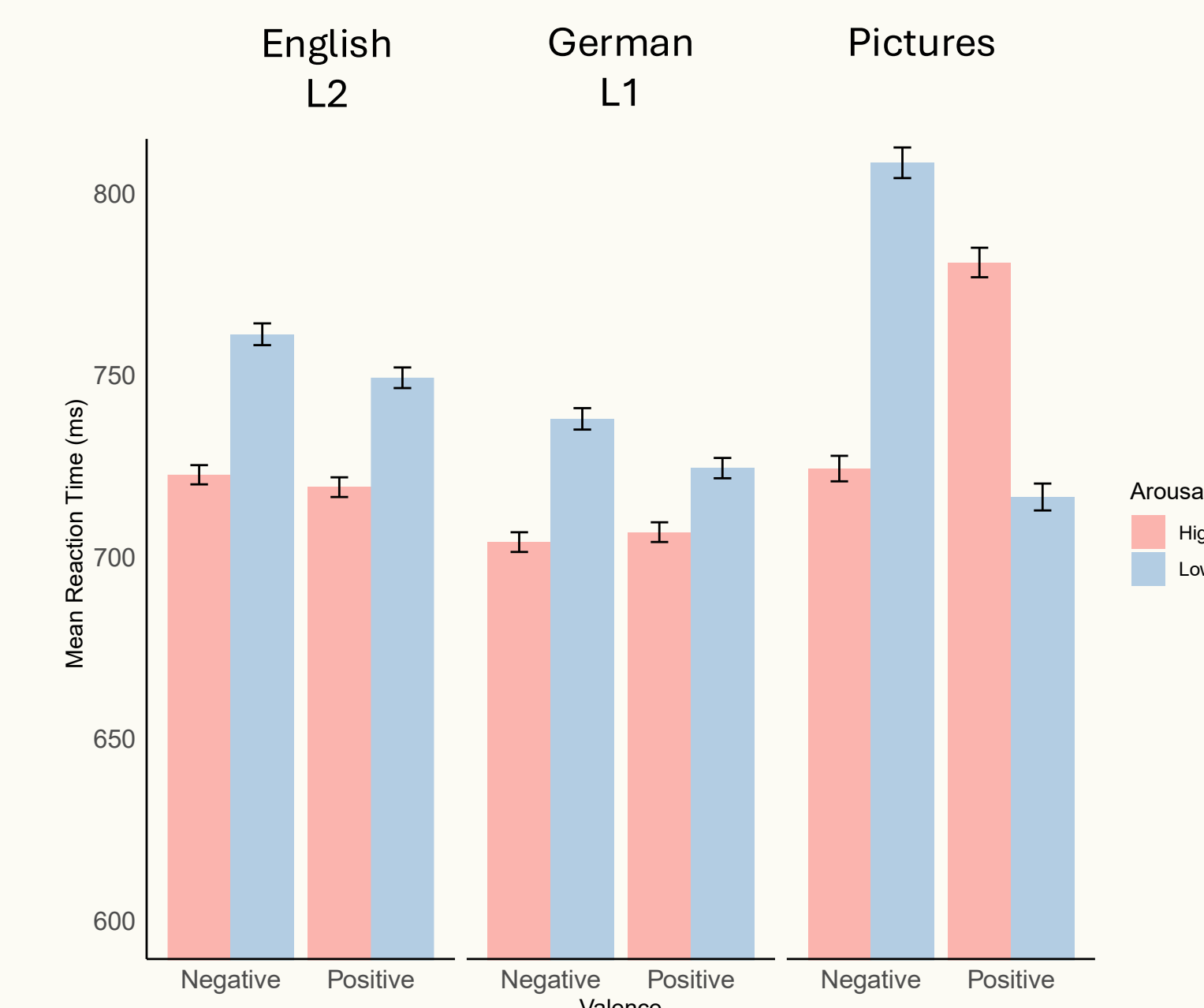
Arousal Task



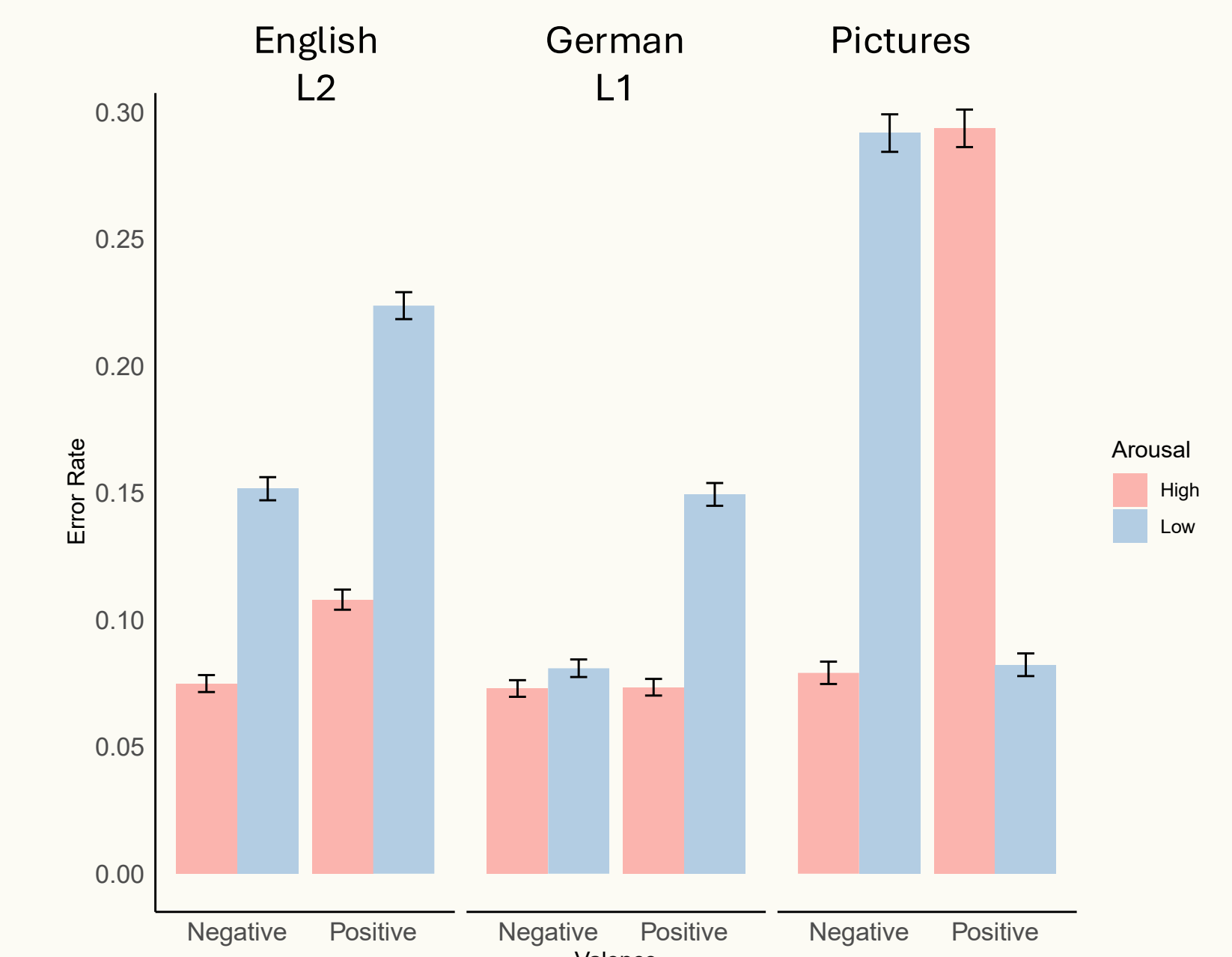
Exp 4:

Results

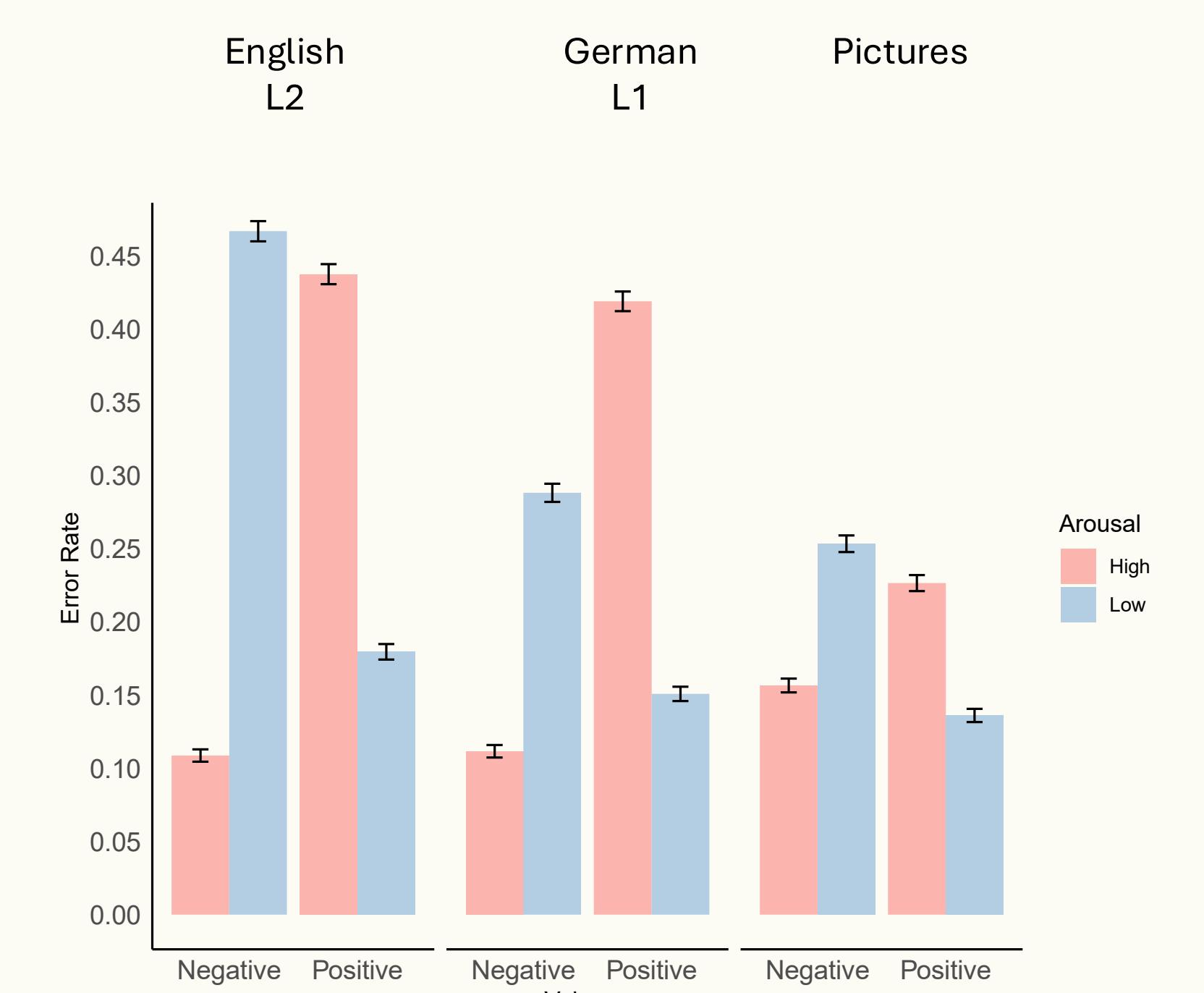
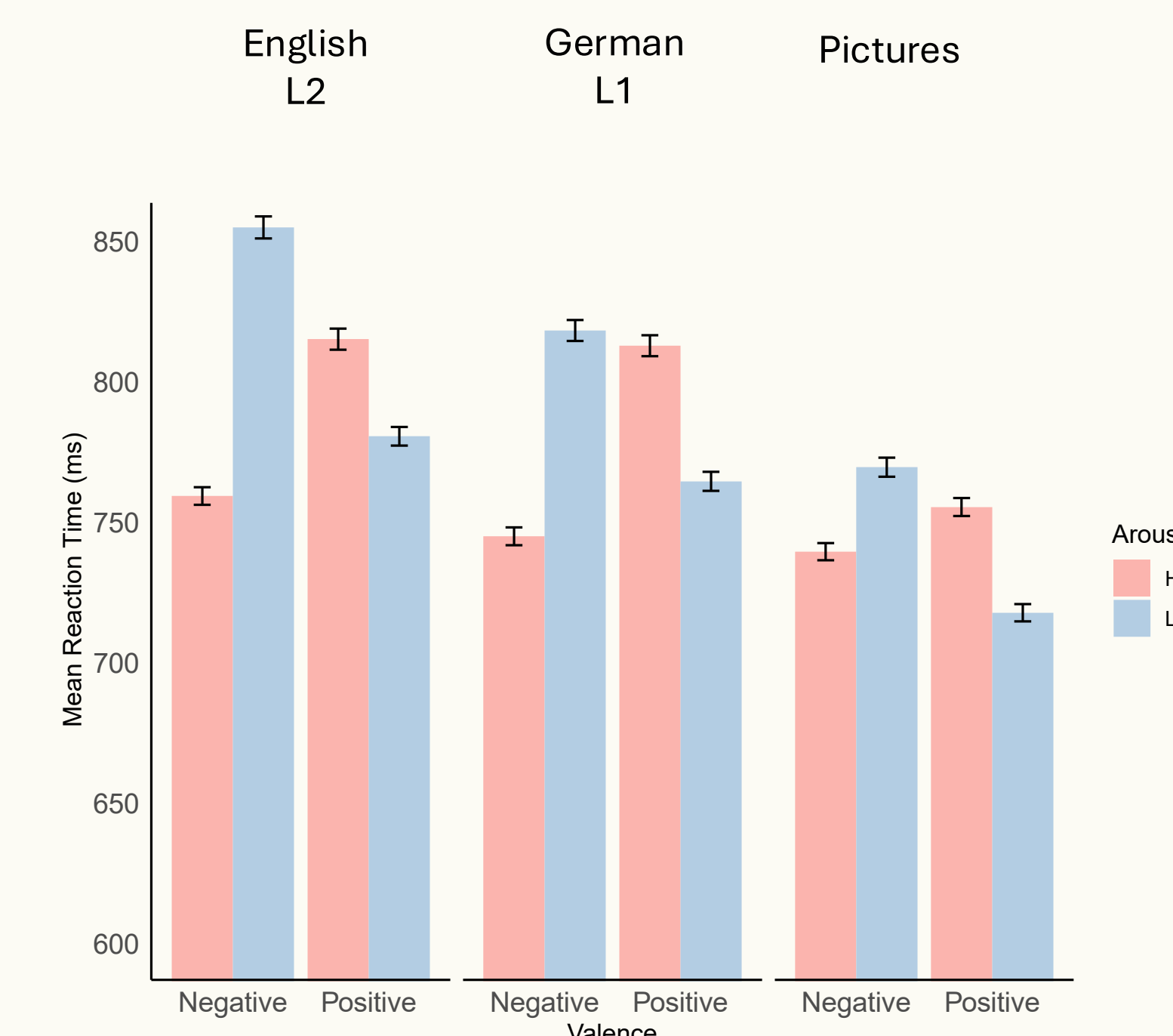
Response Time



Error Rates



Replicated results in separate German-English population (immersed in English-speaking environment), confirming a direct effect of arousal ($b = 30.59, p = .003$), but **no interaction**.



Replicated results in separate German-English population (immersed in English-speaking environment) and found **significant Valence-Arousal Conflict Pattern** ($b = -30.61, p < .001$) across all stimulus types.

This research has been partially funded by the European Union, by the ERC Advanced Grant VALENCE ASYMMETRIES (GA n. 101142133). Views and opinions expressed are however those of the author only and do not necessarily reflect those of the European Union or the European Research Council Executive Agency. Neither the European Union nor the granting authority can be held responsible for them.

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